

CS460 Algorithms, Fall 2025 – Section 03

COURSE INFORMATION

Class Days: T/Th	Instructor: Hyunjong Choi
Class Times: 15:30 – 16:45	Email: hyunjong.choi@sdsu.edu
Class Location: SH 109 (Storm Hall)	Office Location: GMCS 541
Mode: face-to-face for the semester	Office Hours (in-person or Zoom) Tuesday 13:00 – 14:30
Platform: Canvas	Zoom ID: https://SDSU.zoom.us/j/87824147085
Instructional student assistant: Jaden Ong (jong6319@sdsu.edu)	

CATALOG DESCRIPTION

Algorithms for solving frequently occurring problems. Analysis techniques, divide and conquer algorithms with applications, graph problems, and greedy algorithms. Introduction to NP-complete problems.

- **Prerequisites:** CS 210 (Data Structures), Math 245 (Discrete Mathematics), and working with the fundamental knowledge of programming, e.g., C, C++, Java, Python, etc will be plus.

COURSE MATERIALS

Materials	Required or optional	Where and how it can be obtained
Cormen, T. H., Leiserson, C. E., Rivest, R. L., & Stein, C. (2009). <i>Introduction to Algorithms 3rd edition</i> . MIT press.	Optional	https://ebookcentral.proquest.com/lib/sdsu/detail.action?docID=3339142

COURSE DESIGN: MAJOR ASSIGNMENTS AND ASSESSMENTS

Week	Date	Topics	Assignment
1, 2	8/26(Tu)	Introduction of algorithms	
	8/28(Th), 9/2(Tu)	Insertion sort and merge sort	
2, 3	9/4(Th)	Growth of functions	P1
	9/9(Tu), 9/11(Th)	Divide-and-Conquer: maximum-subarray problem, matrix multiplication problem	C1
4, 5	9/16(Tu), 9/18(Th)	Recurrence equations: Substitution method, recursion-tree method, Master theorem	
	9/23(Tu)	Midterm 1 review	
	9/25(Th)	Midterm 1	

Week	Date	Topics	Assignment
6	9/30(Tu), 10/2(Th)	Sorting algorithms: quick sort, heap sort	
7	10/7(Tu)	Sorting algorithms: linear-time sorting	P2
7, 8	10/9(Th), 10/14(Tu), 10/16(Th)	Dynamic programming: rod-cutting problem, matrix-chain multiplication, longest common subsequence	C2
9	10/21(Tu), 10/23(Th)	Greedy algorithm: activity-selection problem, Huffman codes and knapsack problem	
10	10/28(Tu)	Midterm 2 review	
	10/30(Th)	Midterm 2	
11	11/4(Tu)	Graph algorithms: breadth-first search(BFS)	
	11/6(Th)	Graph algorithms: depth-first search(DFS)	P3
12	11/11(Tu), 11/13(Th)	Advanced Graph algorithms: topological sort and strongly connected components	C3
13	11/18(Tu), 11/20(Th)	Minimum spanning tree: Kruskal and Prim algorithms	
14	11/25(Tu), 11/27(Th)	Thanksgiving week	
14	12/2(Tu), 12/4(Th)	Single-source shortest paths: Bellman-Ford algorithm, Dijkstra's algorithm	
16	12/9(Tu), 12/11(Th)	NP-completeness problem, Final review	
	12/18(Th)	Final exam	

COURSE SCHEDULE

Date	Activity
9/25/2025, Thursday, in class	First midterm exam (closed book, 1 cheat paper)
10/30/2025, Thursday, in class	Second midterm exam (closed book, 1 cheat paper)
12/18/2025, Thursday, 13:00 – 15:00	Final exam (closed book, 3 cheat papers)

* Cheat papers: allowed to be double-sided on a letter-sized paper and can be either handwritten or printed.

GRADING POLICIES

Item	Weight	Content
Paper-based assignments (P)	20%	A set of written questions to be completed and submitted.
Coding assignments (C)	20%	Assignments that require Python code to implement specific algorithms including problems to prepare for live coding interviews.
In-lecture / after-class Quizzes	5%	Short quizzes conducted after lectures.

Item	Weight	Content
Midterm exams	30%	Two midterm exams, each worth 15%.
Final exam	25%	Comprehensive exam covering the entire course content.

- Assessments: Paper-based + programming assignments + mid-term exams + final exam
- Quizzes: Popup quizzes with **NO makeup** unless notify before the class.
- For the paper-based assignments, your work needs to be submitted via Canvas (connected to the Gradescope).
- For the programming assignments, a python code file needs to be submitted via Canvas (connected to the Gradescope), A template will be provided via Canvas.
- Late submission penalty for all assignments: **-5 points per day** if it passed the deadline. Make sure to **email me before the deadline** if you are in special situations that prevent you from submitting on time.

GRADING SCALE

Score	Grade
93 ≤	A
90 ≤	A-
87 ≤	B+
83 ≤	B
80 ≤	B-
77 ≤	C+
73 ≤	C
70 ≤	C-
67 ≤	D+
63 ≤	D
60 ≤	D-
0	F

STUDENT LEARNING OUTCOMES

Learning Outcome	Course Activity	Assessment Strategy
Can explain the domains where different algorithms apply and can list the real word problems that each algorithm solves.	In-class quizzes and small multi-choice questions in assignments.	Assessed by the scores in quizzes and assignments.

Learning Outcome	Course Activity	Assessment Strategy
Can use the asymptotic notations (big O etc.) to describe the time and space performance of algorithms and can clearly describe their mathematical meanings.	Questions in assignments that ask students to analyze the complexity of algorithms, and to prove it mathematically.	Assessed by the correctness of the answers, and the clearness of the proof.
Can use pseudocode and Python programming language to implement the algorithms covered and related.	Some questions in exams and assignments ask students to fill up the missing pseudocode in algorithms.	Assessed by the correctness and efficiency of the code. Notation conventions should be followed.
Can propose multiple solutions to complex problems, and then choose the optimal one based on a thorough analysis of their performances	In exams and assignments, students are asked to provide several different solutions to one complex problem, starting from naïve ones to advanced ones.	Assessed by the comprehensiveness of the solutions proposed, and the degree of details in implementation and comparison.
Can describe the relationships between different algorithm families and explain the internal logic of these algorithms are connected.	In-class quizzes where students are asked to explain how problems are different and similar to each other, and why certain algorithms are applied.	Assessed by the scores in quizzes.

COMMUNICATION

My SDSU Gmail address hyunjong.choi@sdsu.edu will be used for all communications. I will use Canvas message to send you important notes, updates, and reminders about class assignments and events. But please **DO NOT** use Canvas message to contact me if you have questions to ask or to discuss. Please **DO** use SDSU email to reach me. For more information, please see [Student Official Email Address Use Policy here](#).

My preferred gender pronouns are he/him. Class rosters are provided to the instructor with the student's legal name. I will gladly honor your request to address you by an alternate name and/or gender pronoun. Please advise me of this early in the semester so that I may make appropriate changes to my records. I prefer to be addressed by Dr. or Prof. Choi. You are welcome to let me know how you prefer to be addressed.

TECHNOLOGY

Canvas (sdsu.instructure.com) will be the main platform for publishing course materials, quiz & assignment questions. The course content will be organized using the “Modules” function of Canvas.

MEDICAL-RELATED ABSENCES

- University policy instructs students to contact their professor/instructor/coach in the event they need to miss class due to an illness, injury, or emergency. All decisions about the impact of an absence, as well as any arrangements for making up work, rest with the instructors.
- If a student misses class because of COVID-19, either because they have been diagnosed and are quarantined or are required to isolate and would like to request a class excuse letter, the student should send an email to vpsafrontdesk@sdsu.edu to notify the university. Student Affairs and Campus Diversity will initiate the process for absent letters to be sent to course instructors, Assistant Deans, and the Provost. Medical documentation may be required prior to the letter being issued.
- [Student Health Services](#) (SHS) does not provide medical excuses for short-term absences due to illness or injury. When a medical-related absence persists beyond five days, SHS will work with students to provide appropriate documentation.
- When a student is hospitalized or has a serious, ongoing illness or injury, SHS will, at the student's request and with the student's consent, communicate with the student's instructors via the Vice President for Student Affairs and Campus Diversity and may communicate with the student's Assistant Dean and/or the [Student Ability Success Center](#).

FINDING HELP ON CAMPUS

Need help finding an advisor, tutor, counselor, or require emergency economic assistance? The [SDSU Student Success Help Desk](#) is here for you. Student assistants are available via Zoom **Monday through Friday, 9:00 AM to 4:30 PM** to help you find the office or service that can best assist with your particular questions or concerns.

- CAL Student Success Center: <https://cal.sdsu.edu/student-resources/student-success>
- College of Education Student Success Center: <https://education.sdsu.edu/oss>
- Center for Student Success in Engineering: <https://csse.sdsu.edu/>
- CoS Student Success Center: <https://cossuccess.sdsu.edu/>
- FSB Student Success Center: <https://business.sdsu.edu/undergrad/advising>
- HHS Advisors: <https://chhs.sdsu.edu/student-resources/advising/>
- IVC Student Success and Retention: https://ivcampus.sdsu.edu/student_affairs/retention
- PSFA Advisors: https://psfa.sdsu.edu/resources/student_advisors

ACADEMIC HONESTY

Students are required to read and adhere to the **SDSU Student Conduct-Related Policies and Guidelines** https://newscenter.sdsu.edu/student_affairs/srr/conduct.aspx

Cheating: Cheating is defined as the act of obtaining or attempting to obtain credit for academic work by the use of dishonest, deceptive, or fraudulent means. Examples of cheating include, but are not limited to:

- Copying, in part or in whole, from another's test or other examination
- Discussing answers or ideas relating to the answers on a test or other examination without the permission of the instructor
- Obtaining copies of a test, an examination, or other course material without the permission of the instructor: Audiotaping of lectures may be done only with the advance permission of the instructor. Permission will be granted to record with the understanding that the recording will be used for the sole purposes of assisting students with classroom notes and erased at the end of the semester. Recorded lectures cannot be posted online. Any other use by students will be considered an infraction of this policy.
- Using notes, cheat sheets, or other devices considered inappropriate under the prescribed testing condition

- Collaborating with another or others in work to be presented without the permission of the instructor
- Falsifying records, laboratory work, or other course data
- Submitting work previously presented in another course, if contrary to the rules of the course
- Altering or interfering with the grading procedures
- Plagiarizing, as defined
- Knowingly and intentionally assisting another student in any of the above

Plagiarism: Plagiarism is defined as the act of incorporating ideas, words, or specific substance of another, whether purchased, borrowed, or otherwise obtained, and submitting same to the university as one's own work to fulfill academic requirements without giving credit to the appropriate source. Plagiarism shall include but not be limited to:

- Submitting work, either in part or in whole, completed by another
- Omitting footnotes for ideas, statements, facts, or conclusions that belong to another
- Omitting quotation marks when quoting directly from another, whether it be a paragraph, sentence, or part thereof
- Close and lengthy paraphrasing of the writings of another
- Submitting another person's artistic works, such as musical compositions, photographs, paintings, drawings, or sculptures
- Submitting as one's own work papers purchased from research companies

Disciplinary Action: Cheating and plagiarism in connection with an academic program at the university may warrant two separate and distinct courses of disciplinary action that may be applied concurrently in response to a violation of this policy:

- Academic sanctions, such as grade modifications
- Punitive sanctions, such as probation, suspension, or expulsion.

Academic sanctions are concerned with the student's grades and are the responsibility of the instructor involved (may receive a zero grade for exam or assignment). Punitive sanctions are concerned with the student's records and status on campus and shall be the responsibility of the university president or designated representative. The Coordinator of Judiciary Procedures shall be the president's representative in matters of student discipline.

Unauthorized recording or dissemination of virtual course instruction or materials by students, especially with the intent to disrupt normal university operations or facilitate academic dishonesty, is a violation of the Student Conduct Code. This includes posting exam problems or questions to online platforms. Violators may be subject to discipline.

The California State University system requires instructors to report all instances of academic misconduct to the Center for Student Rights and Responsibilities. Academic dishonesty will result in disciplinary review by the University and may lead to probation, suspension, or expulsion. Instructors may also, at their discretion, penalize student grades on any assignment or assessment discovered to have been produced in an academically dishonest manner.

CLASSROOM CONDUCT STANDARDS

SDSU students are expected to abide by the terms of the [Student Conduct Code](#) in classrooms and other instructional settings. Violation of these standards will result in referral to appropriate campus authorities. Prohibited conduct includes:

- Willful, material, and substantial disruption or obstruction of a University-related activity, or any on-campus activity.

- Participating in an activity that substantially and materially disrupts the normal operations of the University or infringes on the rights of members of the University community.
- Unauthorized recording, dissemination, or publication (including on websites or social media) of lectures or other course materials.
- Conduct that threatens or endangers the health or safety of any person within or related to the University community, including:
 1. Physical abuse, threats, intimidation, or harassment.
 2. Sexual misconduct.

ACCOMMODATIONS

SDSU via the [Student Ability Success Center](#) (SASC) provides accommodations for students with documented disabilities or medical conditions covered under the Americans with Disabilities Act (ADA). In keeping with current public health guidance, I cannot provide arrangements to students without an ADA-qualified disability or medical condition.

If you are a student with a disability and are in need of accommodations for this class, please contact the Student Ability Success Center at sascinfo@sdsu.edu (or go to sdsu.edu/sasc) as soon as possible. Please know accommodations are not retroactive, and I cannot provide accommodations based upon disability until I have received an accommodation letter from the Student Ability Success Center. SASC registration and accommodation approvals may take up to 10-14 business days, so please plan accordingly.

STUDENT PRIVACY AND INTELLECTUAL PROPERTY

The [Family Educational Rights and Privacy Act](#) (FERPA) mandates the protection of student information, including contact information, grades, and graded assignments. I will use Canvas to communicate with you, and I will not post grades or leave graded assignments in public places. Students will be notified at the time of an assignment if copies of student work will be retained beyond the end of the semester or used as examples for future students or the wider public. Students maintain intellectual property rights to work products they create as part of this course unless they are formally notified otherwise.

RELIGIOUS OBSERVANCES

According to the University Policy File, students should notify instructors of planned absences for religious observances by the end of the second week of classes.

ACADEMIC SUPPORT SERVICES

A complete list of all academic support services—including the [Writing Center](#) and [Math Learning Center](#)—is available on the Student Affairs' [Academic Success](#) website. Counseling & Psychological Services (619-594-5220, sdsu.edu/cps) offers a range of psychological services for students. Emergency support is available after hours at the same phone number. The San Diego Access and Crisis Line can also be accessed 24 hours/day (1-888-724-7240).

SEXUAL VIOLENCE / TITLE IX MANDATED REPORTING

As an instructor, one of my responsibilities is to help create a safe learning environment on our campus. I am a mandated reporter in my role as an SDSU employee. It is my goal that you feel able to share information related to your life experiences in classroom discussions, in your written work, and in our one-on-one meetings. I will seek to keep the information you share private to the greatest extent possible. However, I am required to share information regarding sexual violence on SDSU's campus with the Title IX coordinator, Gail Mendez (619-594-6464). She (or her

designee) will contact you to let you know about accommodations and support services at SDSU and possibilities for holding accountable the person who harmed you. Know that you will not be forced to share information you do not wish to disclose and your level of involvement will be your choice. If you do not want the Title IX Officer notified, instead of disclosing this information to your instructor, you can speak confidentially with the following people on campus and in the community. They can connect you with support services and discuss options for pursuing a University or criminal investigation. Sexual Violence Victim Advocate (619-594-0210) or Counseling and Psychological Services (619-594-5220, psycserv@sdsu.edu). For more information regarding your university rights and options as a survivor of sexual misconduct or sexual violence, please visit titleix.sdsu.edu.

SDSU ECONOMIC CRISIS RESPONSE TEAM

If you or a friend are experiencing food or housing insecurity, technology concerns, or any unforeseen financial crisis, it is easy to get help! Visit sdsu.edu/ecrt for more information or to submit a request for assistance.

SDSU's Economic Crisis Response Team (ECRT) aims to bridge the gap in resources for students experiencing immediate food, housing, or unforeseen financial crises that impact student success. Using a holistic approach to well-being, ECRT supports students through crisis by leveraging a campus-wide collaboration that utilizes on- and off-campus partnerships and provides direct referrals based on each student's unique circumstances. ECRT empowers students to identify and access long-term, sustainable solutions in an effort to successfully graduate from SDSU. Within 24 to 72 hours of submitting a referral, students are contacted by a member of ECRT and are quickly connected to the appropriate resources and services.

For students who need assistance accessing technology for their classes, visit our ECRT website (sdsu.edu/ecrt) to be connected with the SDSU library's technology checkout program. The technology checkout program is available to both SDSU and Imperial Valley students.

LAND ACKNOWLEDGEMENT

For millennia, the Kumeyaay people have been a part of this land. This land has nourished, healed, protected and embraced them for many generations in a relationship of balance and harmony. As members of the San Diego State University community, we acknowledge this legacy. We promote this balance and harmony. We find inspiration from this land, the land of the Kumeyaay.

DIVERSITY AND INCLUSION

Please remember that groups are only effective if everyone treats each other with respect. You are encouraged to communicate your thoughts, but please also allow others in your group to also express their thoughts. You will be surprised how much you can learn by mutual respect of each other's ideas, even if you are knowledgeable about a subject. Remember that you learn best by teaching a subject. Try to be receptive to constructive criticism, and open about accepting mistakes. Please come to the class prepared and take ownership of your group's success.